LESSON PLAN

EHM (28)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Riveting and De-riveting: Types of rivets, rivet parameters calculation, standard riveting procedure, de-riveting of loose rivets, heat treatment
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Riveting and De-riveting: Types of rivets, rivet parameters calculation, standard riveting procedure, de-riveting of loose rivets, heat treatment of rivets, inspection procedure.
5. Specific Objectives	28.1.3.3 Trainee will be able to explain Riveting and De-riveting, Types of rivets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of rivet parameter calculation and standard riveting procedure on practice panel; Safety: Eye protection, fire watch for heat treatment; Supervision: Instructor monitors tool usage and heat application.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of riveting in aircraft structures?
(b) Name two common types of rivet heads used in aviation.
(c) Why is heat treatment required for certain rivet alloys?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification of rivet types • Rivet parameter calculations • Standard riveting procedure • De-riveting loose rivets • Heat treatment of rivets	L & Ds	PPT, IB	15 min
(b)	• Inspection procedure for riveted joints • Common defects in riveting • Safety precautions during heat treatment • Tool maintenance and storage • Quality control checks	L & Ds	PPT, PDF	20 min
(c)	• Review of rivet head deformation • Correct torque application • Cooling rates for heat treated rivets • Documentation of rivet installation • Troubleshooting installation issues	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between solid and blind rivets.
(b) Describe the steps involved in the standard riveting procedure.
(c) What are the critical parameters to check during rivet inspection?

12. Home Assignment and Follow Up	Prepare a brief report comparing the riveting and patch repair techniques covered in LP 05 and LP 06, highlighting key differences in application.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Sheet metal repair - patch repair, stop drilling of cracks.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min